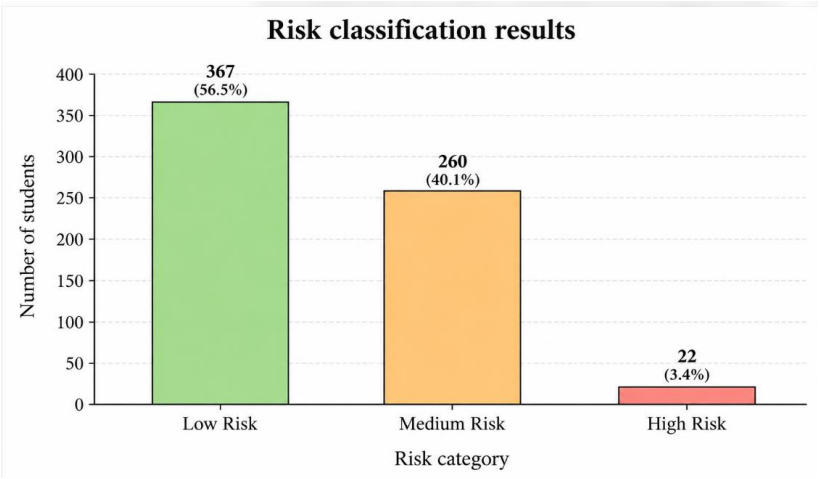
Summary of Results

- Data quality: All 649 student records were processed successfully, with no missing values, invalid values, or duplicate records detected by our cleaning checks.
- Risk classification: The algorithm classified 367 students (56.5%) as Low Risk, 260 (40.1%) as Medium Risk, and 22 (3.4%) as High Risk.
- Prediction accuracy: The algorithm achieved an overall binary pass/fail prediction accuracy of 84.90% using the final grade (`G3`) for evaluation.
- Correctly identified failures: Of the 100 students who actually failed (`G3 < 10`), the algorithm correctly identified 12 as High Risk.
- Missed failures: The algorithm failed to identify 88 of the 100 students who actually failed, indicating that the current risk-scoring rules need improvement.
- False alarms: Ten students were classified as High Risk but actually passed, demonstrating that risk factors do not always lead to academic failure.
- Main conclusion: Although the algorithm achieved relatively high overall accuracy, it identified only 12% of students who actually failed. This suggests that accuracy alone is not sufficient to evaluate its effectiveness.




```
Original records: 649

DATA CLEANING RESULTS
Missing records: 0
Invalid records: 0
Possible duplicates: 0
Cleaned records: 649

RISK CLASSIFICATION RESULTS
Low Risk: 367
Medium Risk: 260
High Risk: 22

EVALUATION RESULTS
True Positives: 12
False Positives: 10
False Negatives: 88
True Negatives: 539
Prediction Accuracy: 84.90%

VERIFICATION
Total classified: 649
Total evaluated: 649
Total cleaned: 649

FIRST 649 STUDENT RESULTS
     failures  studytime  absences    G3  risk_score risk_category  \
0           0          2         4  11.0           0      Low Risk
1           0          2         2  11.0           0      Low Risk
2           0          2         6  12.0           0      Low Risk
3           0          3         0  14.0           0      Low Risk
4           0          2         0  13.0           0      Low Risk
..        ...        ...       ...   ...         ...           ...
644         1          3         4  10.0           1   Medium Risk
645         0          2         4  16.0           0      Low Risk
646         0          2         6   9.0           0      Low Risk
647         0          1         6  10.0           1   Medium Risk
648         0          1         4  11.0           1   Medium Risk

    actual_result
0            Pass
1            Pass
2            Pass
3            Pass
4            Pass
..            ...
644          Pass
645          Pass
646          Fail
647          Pass
648          Pass
```